QUESTION PAPER

Model Paper — 05

Mathematics II

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Time: 3 hours (additional reading time: 10 minutes) **Total marks:** 100

Index number: .....

Important:

- Select **five** questions from Part A and **five** questions from Part B and answer ten questions in all.
- Show the relevant steps and write the correct units when answering.
- Each question carries 10 marks. The marks for each part are shown in brackets.
- Write the answers on separate answer sheets. Where a question says “copy the figure”, copy it on your answer sheet and then answer.
- In constructions use only a straight edge with a cm/mm scale and a compass; the construction lines must be clearly visible.
- Mathematical tables (logarithms, trigonometric ratios) may be used.
- Volume of a right pyramid of base area A and height $h = \frac{1}{3}Ah$. Volume of a right circular cylinder of base radius r and height $h = \pi r^2 h$.

Part A

Answer only five questions.

1. Nimesha and Ruwan each buy the same water pump for their gardens, whose cash price is Rs. 78 000. Nimesha takes a bank loan of Rs. 78 000 and settles it with a single payment at the end of one year, paying Rs. 10 140 as interest. Ruwan buys the pump on hire purchase: he pays 12 equal monthly instalments of Rs. 7 345 at the end of each month. Each instalment includes Rs. 6 500 of the price of the pump, and the total interest of Rs. 10 140 is calculated on the balance of the price still owed (the reducing balance). Show that the annual interest rate of Ruwan's method is higher than that of Nimesha's method, by answering the following.

(i) Find the annual interest rate (as a percentage) of Nimesha's loan. (Marks 2)

(ii) Write the balance of the price owed by Ruwan at the start of the 1st, 2nd and 12th months. (Marks 2)

(iii) The 12 monthly balances form an arithmetic progression. Find the sum of these 12 balances. (Marks 2)

(iv) The interest for one month on a balance of B rupees at r% per annum is $\frac{B \times r}{100 \times 12}$. Using the sum in (iii), find the annual interest rate r of Ruwan's method. (Marks 3)

(v) Hence show that the rate of Ruwan's method is higher than that of Nimesha's method. (Marks 1)

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2. An incomplete table of values for the function $y = f(x)$ for $-6 \le x \le 2$ is given below.

x	−6	−5	−4	−3	−2	−1	0	1	2
y	−7	0	5	8	9	8		0	−7

(i) Using the symmetry of the graph, find the value of y when $x = 0$. (Marks 1)

(ii) Choosing a suitable scale, draw the graph of the function on a sheet of graph paper. (Marks 3)

(iii) (a) Write the x values of the points where the graph cuts the x-axis. (Marks 2)
(b) Using these, write $f(x)$ in the form $y = -(x + p)(x + q)$, where p and q are integers.

(iv) The straight line $y = 4$ cuts the graph at the points A and B. Using the graph, (Marks 4)
(a) write the range of values of x for which $y > 4$,
(b) find the length of AB to the nearest first decimal place.

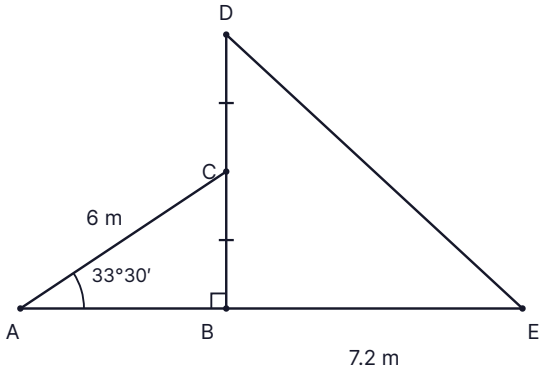
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3. In Kavindu's cricket club grounds, a vertical floodlight pole BD stands on level ground. A, B and E are points on the ground in a straight line, and C is the mid-point of BD. A cable of length 6 m is stretched from A to C, and another cable is stretched from D to E. BE = 7.2 m and ∠CAB = 33°30′. Copy the figure on your answer sheet and mark the given information.

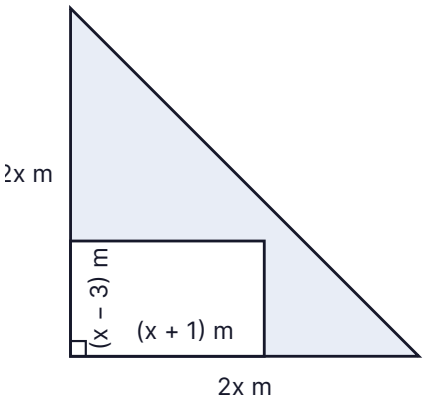
(i) Using the trigonometric tables, find the length of BC. (Marks 3)

(ii) Find the value of tan ∠DEB. (Marks 3)

(iii) Hence show that the angle of inclination of the cable DE (∠DEB) is greater than the angle of inclination of the cable AC (∠CAB) by more than 8°. (Marks 4)



4. Nimesha's garden plot is in the shape of a right-angled isosceles triangle whose two equal sides are each $2x$ metres long. A rectangular vegetable bed of sides $(x + 1)$ m and $(x - 3)$ m is made at the right-angled corner, as shown in the figure. The rest of the plot (shaded) is a lawn of area 100 m^2.



(i) Show that $x^2 + 2x - 97 = 0$. (Marks 3)

(ii) Solve the equation and find the value of x to the nearest whole number. (Take $\sqrt{2} = 1.41$. Note that $\sqrt{392} = 14\sqrt{2}$.) (Marks 4)

(iii) Show that the longest side of the plot is longer than 25 m. (Marks 3)

5. Sachini buys 36 flower pots for her garden. Small pots cost Rs. 65 each and large pots cost Rs. 110 each. She pays Rs. 3 060 in all.

(i) By taking the number of small pots as x and the number of large pots as y, form a pair of simultaneous equations and find the number of pots of each size. (Marks 5)

(ii) Each small pot holds the same number of plants, and each large pot holds 3 more plants than a small pot. The 36 pots together hold 336 plants. Find the number of plants a small pot holds. (Marks 5)

6. The table shows the number of runs scored by Kavindu's cricket club in each of 40 matches.

Runs scored	60 – 69	70 – 79	80 – 89	90 – 99	100 – 109	110 – 119
Number of matches	3	7	9	9	6	6

(i) Find the mean number of runs scored per match. (Marks 3)

(ii) Using the mean, find the expected total number of runs the club will score in a season of 12 matches. (Marks 2)

(iii) A sponsor pays the club Rs. 25 for every run scored, and the club pays Rs. 1 200 per match for the use of the ground. Find the club's expected net income for the 12 matches. (Marks 3)

(iv) Find the minimum possible total number of runs scored in those matches in which the club scored fewer than 80 runs. (Marks 2)

Part B

Answer only five questions.

7. Use only a straight edge with a cm/mm scale and a compass for the constructions below. Show the construction lines clearly.

(i) Construct the line segment AB = 6.4 cm. At A, construct ∠BAP = 60° and ∠BAQ = 90° on the same side of AB, and bisect ∠PAQ. Let the bisector be AX. (Marks 4)

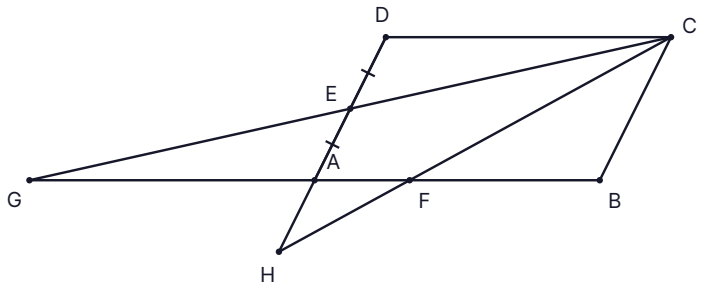
(ii) Mark the point C on AX such that AC = AB, and join BC to complete the isosceles triangle ABC. (Marks 1)

(iii) Construct the perpendicular bisector of AC. With AC as the diameter, construct a semicircle on the side of AC away from B. (Marks 2)

(iv) Construct the line through C parallel to AB, and let it meet the semicircle at D. Join AD. (Marks 2)

(v) Find the size of ∠CAD, giving reasons. (Marks 1)

8. In the parallelogram ABCD, E is the mid-point of AD. CE produced meets BA produced at G. F is the point on AB such that AF = $\frac{1}{3}$ AB. CF produced meets DA produced at H. Copy the figure on your answer sheet.



(i) Show that the triangles AGE and DCE are congruent. (Marks 3)

(ii) Join AC and GD. Show that AGDC is a parallelogram. (Marks 2)

(iii) Show that the triangles AFH and BFC are equiangular. (Marks 3)

(iv) Hence show that AH = $\frac{1}{2}$ AD. (Marks 2)

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9. The cricket club casts solid metal pyramids as prizes. Each pyramid has a square base of side a cm and height $4a$ cm. Six such identical pyramids are melted, without any loss of metal, and recast as one solid right circular cylinder of base radius r cm and height $5r$ cm.

(i) Show that $a^3 = \frac{5\pi}{8} r^3$. (Marks 4)

(ii) When $r = 5.2$ and $\pi = 3.14$, find the value of a^3 using logarithms, and hence find a to the first decimal place. (Marks 3)

(iii) A pyramid of base side a cm and height $4a$ cm is cut out of a solid metal cuboid with a square base of side a cm and height $4a$ cm. Using the value of a^3 found in (ii), find the volume of metal that remains. (Marks 3)

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10. At cricket practice, Kavindu bowls 20 balls on the first day and bowls 5 more balls each day than on the previous day, until he completes his training plan on the 14th day.

(i) Write the number of balls he bowls on the first three days. (Marks 1)

(ii) Find the number of balls he bowls on the 14th day. (Marks 2)

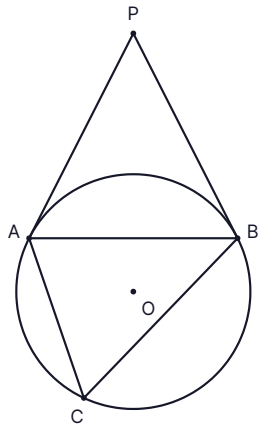
(iii) Find the total number of balls he bowls during the 14 days. (Marks 2)

(iv) Ruwan bowls the same total number of balls in 15 days, bowling 3 more balls each day than on the previous day. Find the number of balls Ruwan bowls on his first day. (Marks 3)

(v) On which day do Kavindu and Ruwan bowl the same number of balls? (Marks 2)

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11. In the figure, O is the centre of the circle. PA and PB are the tangents drawn to the circle from the external point P, touching it at A and B. C is a point on the circle on the opposite side of AB to P. Copy the figure on your answer sheet and join OA and OB.



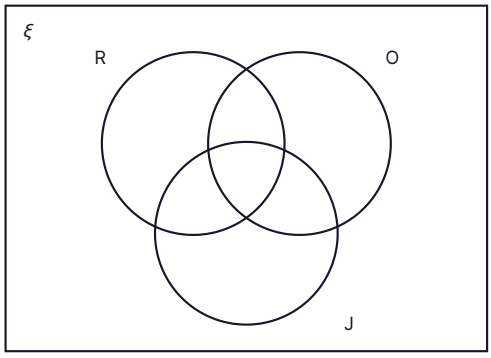
(i) Show that APBO is a cyclic quadrilateral. (Marks 3)

(ii) Show that $\angle APB = 180° - 2 \times \angle ACB$. (Marks 3)

(iii) Name two angles in the figure that are each equal to $\angle PAB$, giving a reason for each. (Marks 4)

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12. 88 customers came to Sachini's plant nursery one day. The universal set ξ is the set of these customers, and R, O and J are the sets of customers who bought roses, orchids and jasmine respectively. 7 customers bought none of the three kinds, and 57 customers bought exactly one kind. Copy the Venn diagram on your answer sheet and enter the information in it.



(i) Find the number of customers who bought two or more kinds. (Marks 3)

(ii) The number of customers who bought all three kinds is half the number who bought exactly two kinds. Find the number who bought all three kinds. (Marks 3)

(iii) 4 customers bought only roses and orchids. The number who bought only orchids and jasmine is twice the number who bought only roses and jasmine. Find these two numbers. (Marks 2)

(iv) 33 customers bought roses and 30 customers bought jasmine. Find the number of customers who bought only orchids. (Marks 2)

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